

## Assignment - 02

### Section - A

1. What is the need for Data Preprocessing?

- Ans: i) Real-world data is often incomplete, noisy and inconsistent.
- ii) Data preprocessing improves data quality, which is crucial for accurate analysis and mining.
- iii) It ensures clean, integrated, consistent & transformed data ready for use in a data warehouse or mining process.

2. Explain different types of data repositories.

- Ans: i) Operational DBs: Used for day-to-day business operations (eg: sales, payroll).
- ii) Data Warehouses: Store large historical data from multiple sources for analysis and reporting.
- iii) Data Marts: Subset of DW focused on a specific business area or department.
- iv) Metadata Repositories: Contain data about data (schemas, structure, rules).
- v) Web Repositories: Store unstructured or semi-structured data from the web.

3. State the different layers of DW.

- Ans. i) Data Source layer  
 ii) Data Staging layer  
 iii) Data Storage layer  
 iv) Data Presentation layer  
 v) MetaData layer.

4. What is the difference b/w DW & DM?

| Ans. Data Warehousing                    | Data Mining                                  |
|------------------------------------------|----------------------------------------------|
| i) Stores and manages data               | Analyzes data to find patterns               |
| ii) Focuses on data storage              | Focus on data analysis                       |
| iii) Supports decision-making indirectly | Supports decision-making directly.           |
| iv) Uses ETL to load data                | Uses algorithms (classification, clustering) |

5. What are the issues in Data Integration?

- Ans. • Schema Integration conflicts  
 • Data Redundancy and Duplications.  
 • Data Value conflicts.  
 • Handling missing or incomplete data.  
 • Maintaining data quality & consistency.



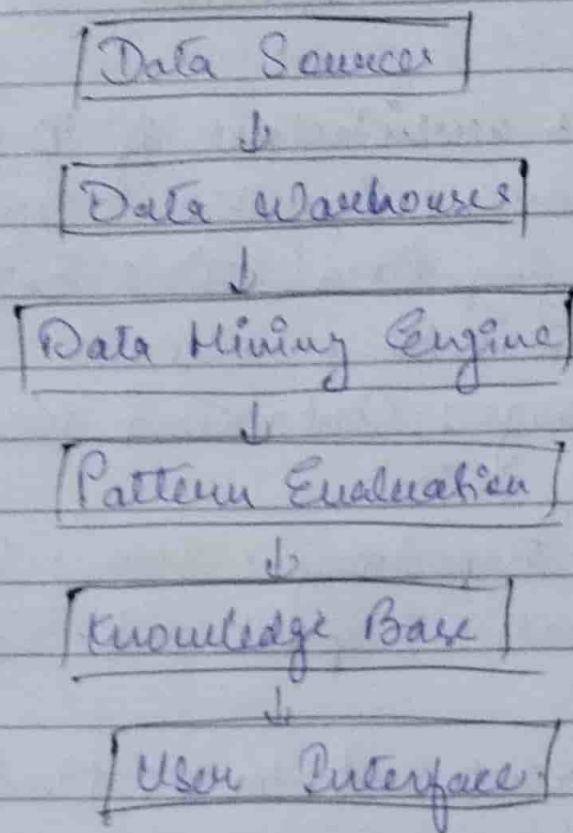
## Section - B.

1. Discuss the architecture of Data Warehouse.

- Ans: i) Data Sources: Operational systems, external data  
ii) Data Staging Area (ETL): Extract, transform & load data into warehouse.  
iii) Data Storage: Centralized data warehouse storing historical data.  
iv) Metadata Repository: Stores info. about data.  
v) Data Marts: Department-specific subsets of data.  
vi) OLAP / Front-end tools: Used by users to analyse & visualize data.

2. Explain the architecture of Data Mining with diagram.

- Ans: i) Data Sources: Databases, data warehouse, web data.  
ii) Data Warehouses: Integrated storage for cleaned data.  
iii) Data Mining Engine: Core component that applies algorithms (classification, clustering, association).  
iv) Pattern Evaluation Module: Evaluates interesting patterns.  
v) Knowledge Base: Stores user knowledge, constraints and rules.  
vi) User Interface: Allows users to interact and visualize results.



3. What are the strategies of Data Reduction?

- Ans: i) Data Cub: Aggregation: Summarize data in multi-dimensional form.
- ii) Dimensionality Reduction: Remove irrelevant or redundant attributes (PCA, feature selection).
- iii) Data Compression: Use encoding to reduce data size.
- iv) Numerosity Reduction: Replace data with models (regression, histograms, clustering).
- v) Data Discretization: Convert continuous data into categorical data.



## Section - C

1. Define KDD. Identify and describe phases of KDD.

Ans. Knowledge Discovery in Databases is a process of extracting useful, hidden, and interesting knowledge from large amounts of data.

Phases:

- i) Data Selection - Choose relevant data from DBs.
- ii) Data Preprocessing - Clean noisy/incomplete data.
- iii) Data Transformation: Convert data into suitable format.
- iv) Data Mining: Apply algos to extract patterns.
- v) Pattern Evaluation: Identify truly useful patterns.
- vi) Knowledge Presentation: Visualize & present the knowledge to the user.

2. Illustrate Data Preprocessing in detail.

Ans. Data Preprocessing prepares raw data for mining & analysis. It involves:

- i) Data Cleaning - Remove noise, handle missing values, correct inconsistencies.
- ii) Data Integration - Merge data from multiple sources.
- iii) Data Transformation - Normalize, scale, or encode data into suitable forms.

- iv) Data Reduction - Reduce data volume while retaining integrity (aggregation, compression, dimensionality reduction).
- v) Data Discretization - Convert continuous data into categorical.

3. Why do we process data? Highlight the importance of each stage in production.

Ans: Raw Data is often incomplete, inconsistent and noisy. Processing makes it clean, consistent and ready for analysis.

Stages and importance :-

- i) Data Collection - Gather correct and sufficient data.
- ii) Data Cleaning - Improves accuracy & reliability.
- iii) Data Integration - Combines data from various sources.
- iv) Data Transformation - Converts data into a standard usable form.
- v) Data Reduction - Reduces data size for faster processing.
- vi) Data Mining - Extracts useful patterns.
- vii) Interpretations & Evaluation - Validates and presents results.